

## SAFETY DATA SHEET

Creation Date 06-October-2009

Revision Date 24-December-2021

Revision Number 5

### 1. Identification

**Product Name** Perchloric acid

**Cat No. :** A229-61LB, A2298LB, A229-S8LB

**CAS-No** 7601-90-3

**Synonyms** Dioxonium perchlorate; Hydronium perchlorate; Perchloric acid solution

**Recommended Use** Laboratory chemicals.

**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Manufacturer**  
Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

**Emergency Telephone Number** CHEMTREC®, Inside the USA: 800-424-9300  
CHEMTREC®, Outside the USA: 001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                             |              |
|-------------------------------------------------------------|--------------|
| <b>Oxidizing liquids</b>                                    | Category 1   |
| <b>Corrosive to metals</b>                                  | Category 1   |
| <b>Acute oral toxicity</b>                                  | Category 4   |
| <b>Skin Corrosion/Irritation</b>                            | Category 1 A |
| <b>Serious Eye Damage/Eye Irritation</b>                    | Category 1   |
| <b>Specific target organ toxicity (single exposure)</b>     | Category 3   |
| Target Organs - Respiratory system.                         |              |
| <b>Specific target organ toxicity - (repeated exposure)</b> | Category 2   |
| Target Organs - Thyroid.                                    |              |
| <b>Physical Hazards Not Otherwise Classified</b>            | Category 1   |
| Risk of explosion if heated under confinement               |              |

#### Label Elements

**Signal Word**

Danger

**Hazard Statements**

May cause fire or explosion; strong oxidizer  
May be corrosive to metals  
Harmful if swallowed  
Causes severe skin burns and eye damage  
May cause respiratory irritation  
May cause damage to organs through prolonged or repeated exposure  
Risk of explosion if heated under confinement

**Precautionary Statements****Prevention**

Wear protective gloves/protective clothing/eye protection/face protection  
Wear fire/flammable resistant/retardant clothing  
Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Keep cool. Protect from sunlight  
Keep/Store away from clothing/combustible materials  
Take any precaution to avoid mixing with combustibles  
Keep only in original container  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area

**Response**

Explosion risk in case of fire  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
IF ON CLOTHING: Rinse immediately contaminated clothing and skin with plenty of water before removing clothes  
Immediately call a POISON CENTER/doctor  
Rinse mouth  
Do NOT induce vomiting  
Rinse skin with water/shower  
Wash contaminated clothing before reuse  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion  
Absorb spillage to prevent material damage

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed  
Store in corrosive resistant polypropylene container with a resistant liner  
Store in a dry place

**Disposal**

Dispose of contents/container to an approved waste disposal plant

### 3. Composition/Information on Ingredients

| Component       | CAS-No    | Weight % |
|-----------------|-----------|----------|
| Perchloric acid | 7601-90-3 | 60-70    |

|       |           |       |
|-------|-----------|-------|
| Water | 7732-18-5 | 30-40 |
|-------|-----------|-------|

#### 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                                                |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.                                                                                                                                                                                                                              |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.                                                                                                                                                                                              |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.                                                                                                                                        |
| <b>Inhalation</b>                      | If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately. |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an unconscious person. Call a physician immediately.                                                                                                                                                                                           |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated                                |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                                          |

#### 5. Fire-fighting measures

|                                         |                                                                   |
|-----------------------------------------|-------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                          |
| <b>Flash Point</b>                      | 113 °C / 235.4 °F                                                 |
| <b>Method -</b>                         | No information available                                          |
| <b>Autoignition Temperature</b>         | No information available                                          |
| <b>Explosion Limits</b>                 |                                                                   |
| <b>Upper</b>                            | No data available                                                 |
| <b>Lower</b>                            | No data available                                                 |
| <b>Oxidizing Properties</b>             | Oxidizer                                                          |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                                          |

#### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes. Oxidizer: Contact with combustible/organic material may cause fire. May ignite combustibles (wood paper, oil, clothing, etc.).

#### Hazardous Combustion Products

Hydrogen chloride gas.

#### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

#### NFPA

Health  
3Flammability  
0Instability  
3Physical hazards  
OX

## 6. Accidental release measures

|                                             |                                                                                                                                                               |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak. |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                                                  |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Sweep up and shovel into suitable containers for disposal.           |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                      |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. Keep away from clothing and other combustible materials. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Do not store near combustible materials. Corrosives area. Incompatible Materials. Strong oxidizing agents. Finely powdered metals. Organic materials. Amines. Alcohols. Strong reducing agents. Combustible material.       |

## 8. Exposure controls / personal protection

|                            |                                                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b> | This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies. |
|----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|

OSHA - Occupational Safety and Health Administration

### Engineering Measures

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location. Ensure adequate ventilation, especially in confined areas.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection**  
**Hand Protection**

Goggles  
Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Butyl rubber   | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### Respiratory Protection

No protective equipment is needed under normal use conditions.

**Recommended Filter type:** Particulates filter conforming to EN 143 or Acid gases filter Type E Yellow conforming to EN14387

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                              |
|----------------------------------------|------------------------------|
| Physical State                         | Liquid                       |
| Appearance                             | Colorless                    |
| Odor                                   | Strong                       |
| Odor Threshold                         | No information available     |
| pH                                     | 0.1 @ 20°C                   |
| Melting Point/Range                    | -18 °C / -0.4 °F             |
| Boiling Point/Range                    | 203 °C / 397.4 °F @ 760 mmHg |
| Flash Point                            | 113 °C / 235.4 °F            |
| Evaporation Rate                       | No information available     |
| Flammability (solid,gas)               | Not applicable               |
| Flammability or explosive limits       |                              |
| Upper                                  | No data available            |
| Lower                                  | No data available            |
| Vapor Pressure                         | 6.8 mmHg @ 25 °C             |
| Vapor Density                          | 3.46                         |
| Specific Gravity                       | 1.66                         |
| Solubility                             | Soluble in water             |
| Partition coefficient; n-octanol/water | No data available            |
| Autoignition Temperature               | No information available     |
| Decomposition Temperature              | No information available     |
| Viscosity                              | 3.5 mPa.s @ 20 °C            |
| Molecular Formula                      | H Cl O <sub>4</sub>          |
| Molecular Weight                       | 100.46                       |

## 10. Stability and reactivity

|                                  |                                                                                                                                    |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                                                |
| Stability                        | Oxidizer: Contact with combustible/organic material may cause fire.                                                                |
| Conditions to Avoid              | Incompatible products. Excess heat. Combustible material.                                                                          |
| Incompatible Materials           | Strong oxidizing agents, Finely powdered metals, Organic materials, Amines, Alcohols, Strong reducing agents, Combustible material |
| Hazardous Decomposition Products | Hydrogen chloride gas                                                                                                              |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                           |
| Hazardous Reactions              | None under normal processing.                                                                                                      |

## 11. Toxicological information

**Acute Toxicity****Product Information**

|             |                                                                               |
|-------------|-------------------------------------------------------------------------------|
| Oral LD50   | Category 4. ATE = 300 - 2000 mg/kg.                                           |
| Dermal LD50 | Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg. |
| Vapor LC50  | Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.    |

**Component Information**

| Component | LD50 Oral | LD50 Dermal | LC50 Inhalation |
|-----------|-----------|-------------|-----------------|
|-----------|-----------|-------------|-----------------|

|       |   |   |   |
|-------|---|---|---|
| Water | - | - | - |
|-------|---|---|---|

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** Causes severe burns by all exposure routes

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component       | CAS-No    | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------|-----------|------------|------------|------------|------------|------------|
| Perchloric acid | 7601-90-3 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Water           | 7732-18-5 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system  
**STOT - repeated exposure** Thyroid

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation: Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. .

**Persistence and Degradability** Soluble in water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN1873  
 Proper Shipping Name PERCHLORIC ACID  
 Hazard Class 5.1  
 Subsidiary Hazard Class 8  
 Packing Group I

### TDG

UN-No UN1873

**Proper Shipping Name** PERCHLORIC ACID  
**Hazard Class** 5.1  
**Subsidiary Hazard Class** 8  
**Packing Group** I

**IATA**

**UN-No** UN1873  
**Proper Shipping Name** PERCHLORIC ACID  
**Hazard Class** 5.1  
**Subsidiary Hazard Class** 8  
**Packing Group** I

**IMDG/IMO**

**UN-No** UN1873  
**Proper Shipping Name** PERCHLORIC ACID  
**Hazard Class** 5.1  
**Subsidiary Hazard Class** 8  
**Packing Group** I

## 15. Regulatory information

**International Inventories**

| Component       | CAS-No    | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------------|-----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Perchloric acid | 7601-90-3 | X   | -    | X    | ACTIVE                                        | 231-512-4 | -      | -   |
| Water           | 7732-18-5 | X   | -    | X    | ACTIVE                                        | 231-791-2 | -      | -   |

| Component       | CAS-No    | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------------|-----------|-------|----------|------|------|------|------|-------|-------|
| Perchloric acid | 7601-90-3 | X     | KE-28137 | X    | X    | X    | X    | X     | X     |
| Water           | 7732-18-5 | X     | KE-35400 | X    | -    | X    | X    | X     | X     |

**Legend:**

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations****Authorisation/Restrictions according to EU REACH**

| Component       | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Perchloric acid | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component       | CAS-No    | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------|-----------|----------|------------------------------|---------------------------|--------------------------------------------|
| Perchloric acid | 7601-90-3 | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Water           | 7732-18-5 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component       | CAS-No    | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------|-----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Perchloric acid | 7601-90-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Water           | 7732-18-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

Regulatory Affairs  
Thermo Fisher Scientific  
Email: EMSDS.RA@thermofisher.com

**Creation Date**

06-October-2009

**Revision Date**

24-December-2021

**Print Date**

24-December-2021

**Revision Summary**

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**